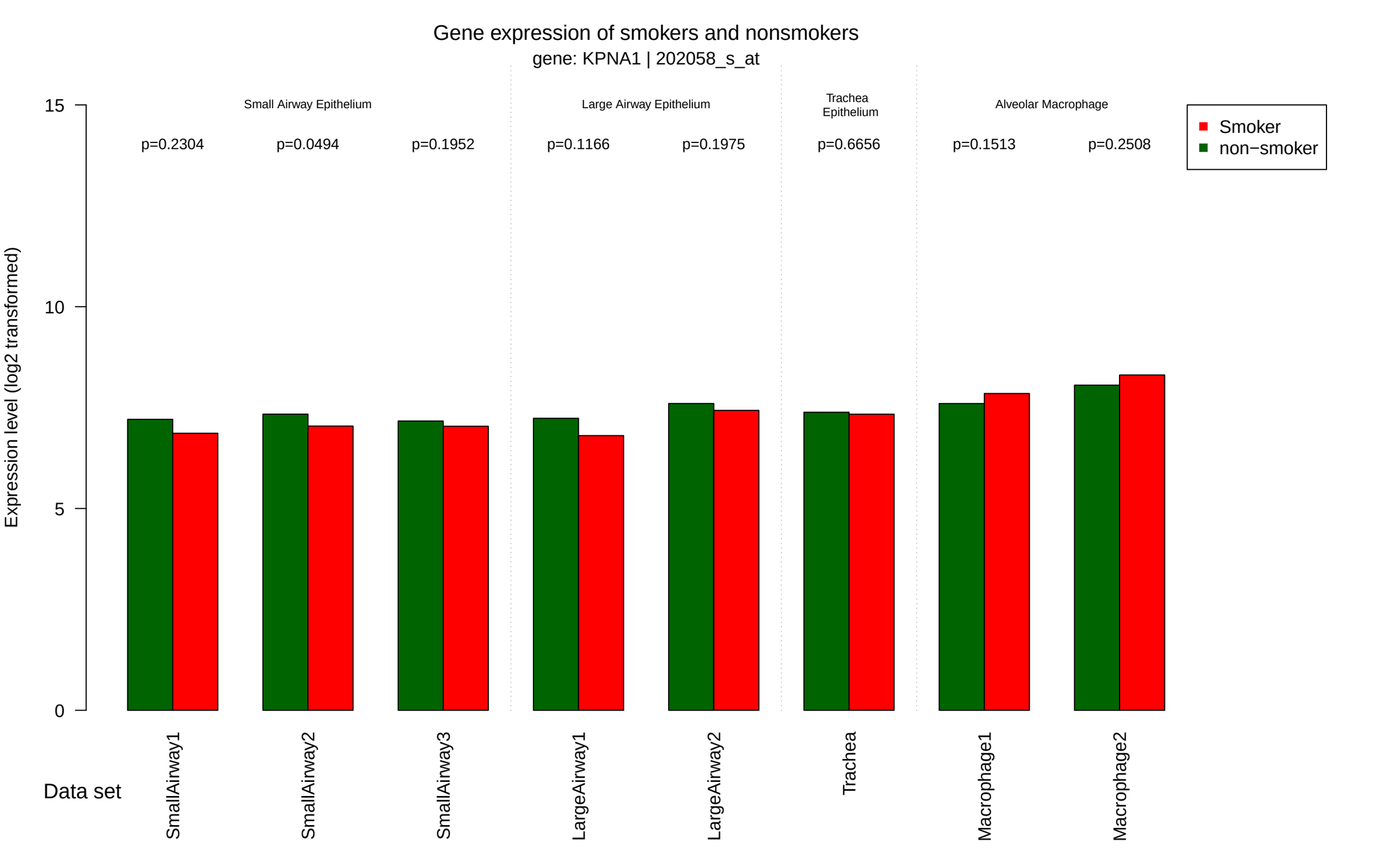
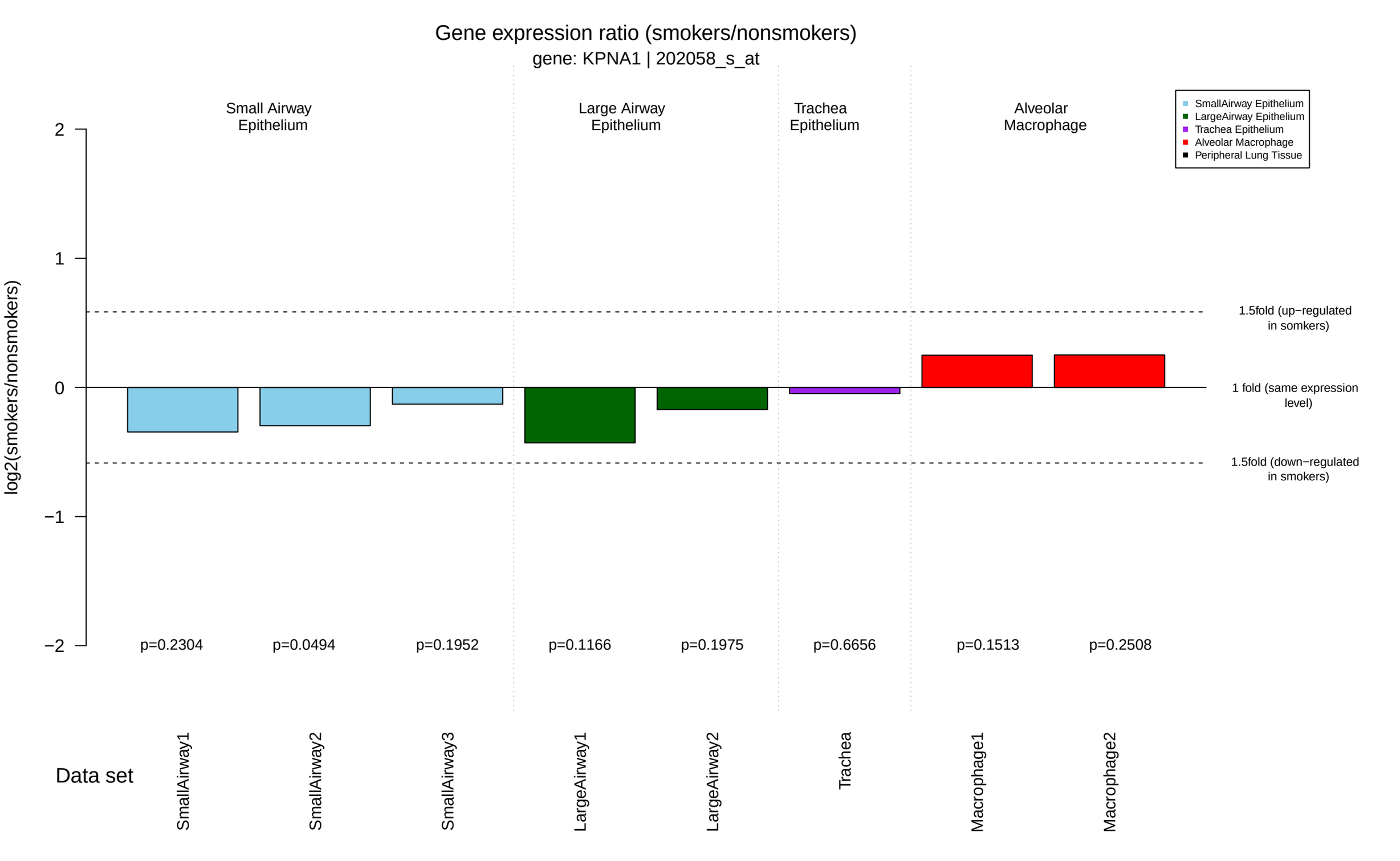
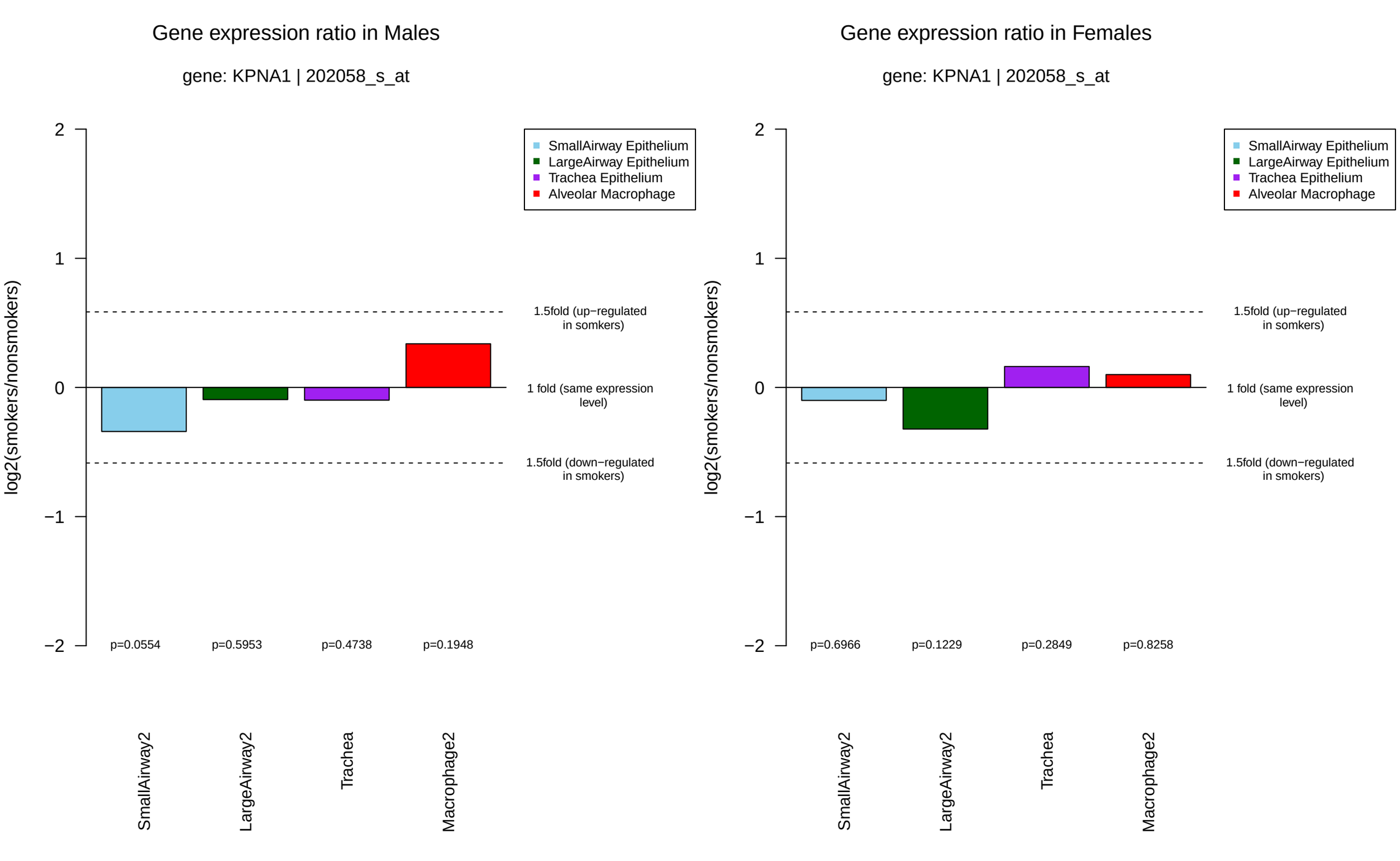
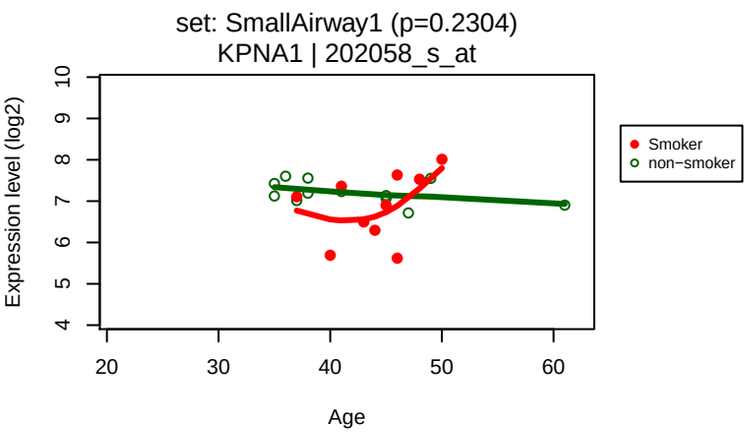
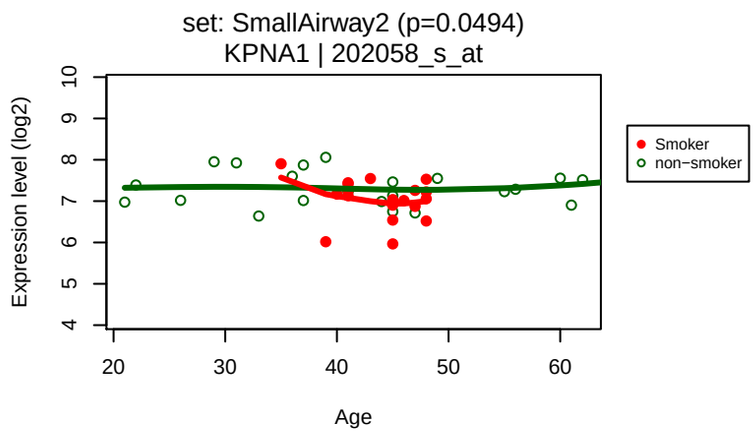
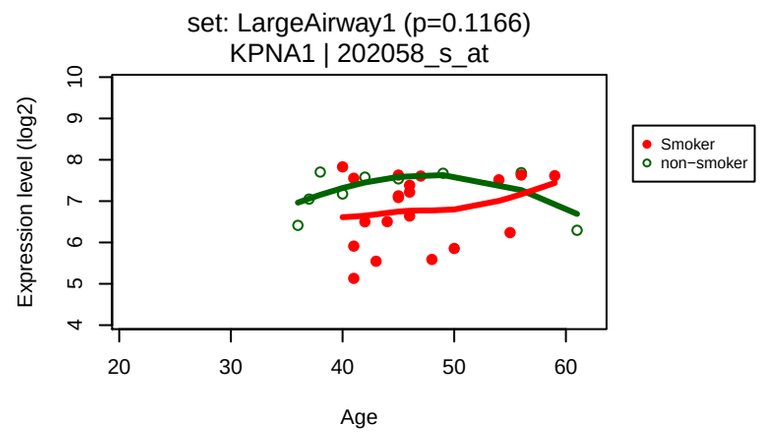

Gene expression of smokers and nonsmokers
gene: KPNA1 | 202058_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2304
p=0.0494
p=0.1952
p=0.1166
p=0.1975
p=0.6656
p=0.1513
p=0.2508
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

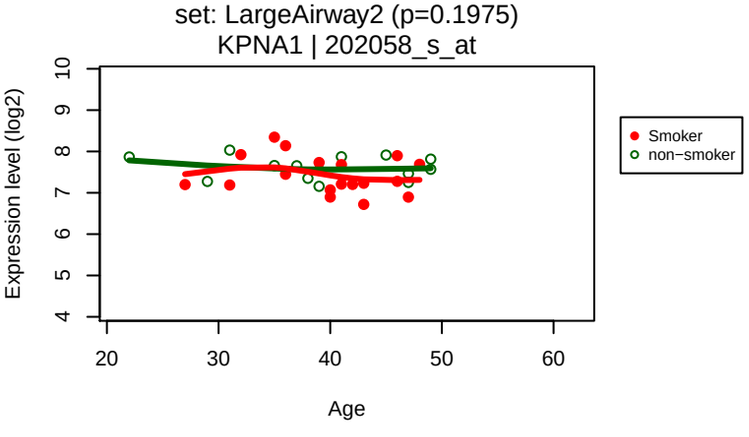

Gene expression ratio (smokers/nonsmokers)
gene: KPNA1 | 202058_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2304
p=0.0494
p=0.1952
p=0.1166
p=0.1975
p=0.6656
p=0.1513
p=0.2508
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

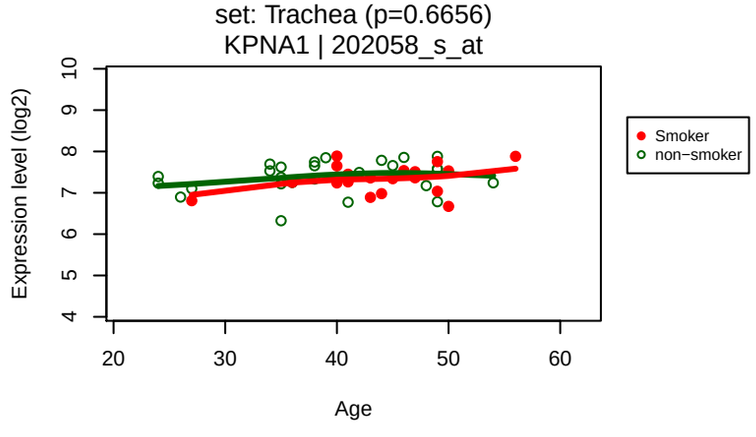

Gene expression ratio in Males
gene: KPNA1 | 202058_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0554
p=0.5953
p=0.4738
p=0.1948
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA1 | 202058_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6966
p=0.1229
p=0.2849
p=0.8258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

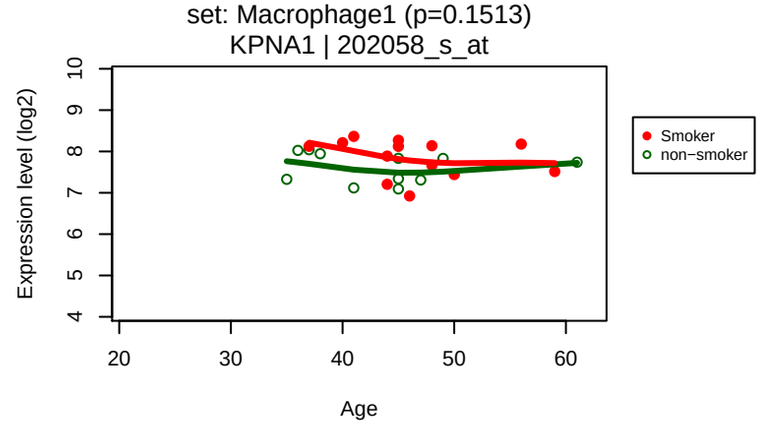

set: SmallAirway1 (p=0.2304)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker

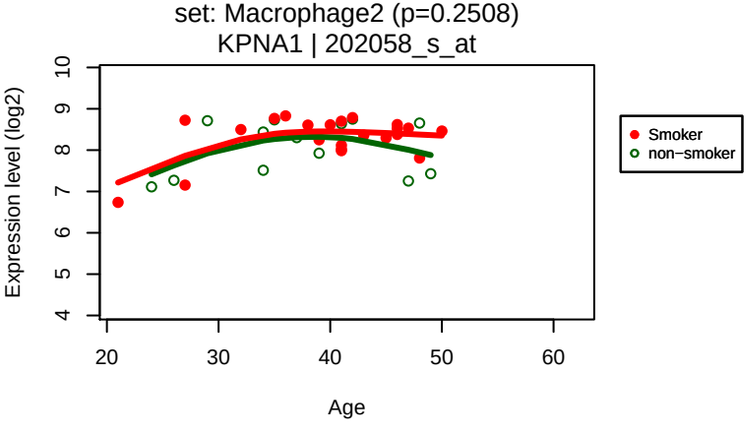

set: SmallAirway2 (p=0.0494)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker

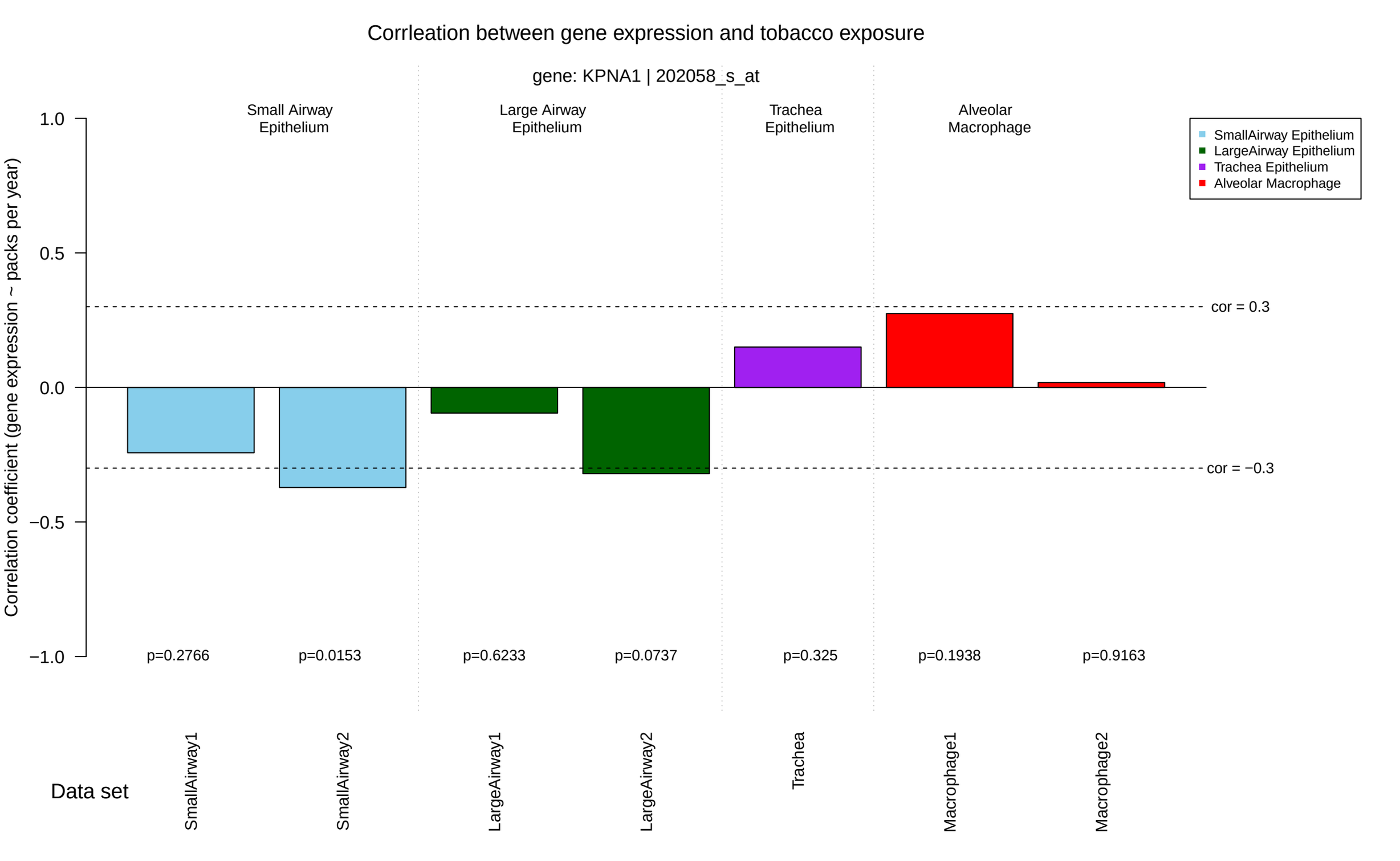

set: LargeAirway1 (p=0.1166)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1975)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6656)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1513)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2508)
KPNA1 | 202058_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA1 | 202058_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2766
p=0.0153
p=0.6233
p=0.0737
p=0.325
p=0.1938
p=0.9163
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2